

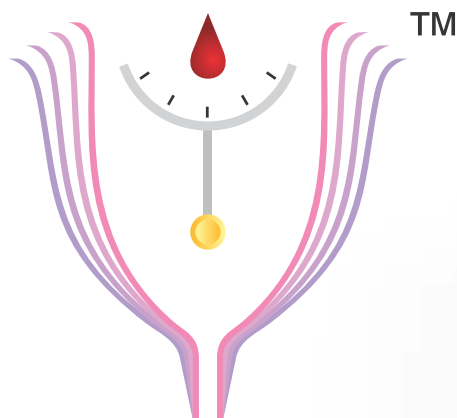


Ministry of Women & Child Development Government of India

PRADHAN MANTRI RASHTRIYA BAL PURASKAR

BAL SHAKTI PURASKAR

(Category-Innovation)



POST PARTUM CUP

A novel way to estimate the obstetric
blood loss accurately and save
Maternal Life in **Golden Hour**

**SAVE MOTHER
&
SAVE MOTHER EARTH**

Project by : Archit Rahul Patil

Guide : Mrs. Sangita Khanna

Principal : Mr. Amit Bhatia Sir,
Kashinath Palod Public School, Jalgaon.





Vivekanand Pratishthan's
Kashinath Palod Public School

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Outward No. :- 392 / 2019-20

Date :- 17 March 2020

AUTHENTICATION CERTIFICATE

This is to authenticate and certify that, Master. **ARCHIT RAHUL PATIL** Son of **RAHUL DATTATRAY PATIL** is a bonafied student of our school studying in Class VIII – Edison. He was admitted to school on 15 June 2009. Admission number 1460. His date of birth in school record is 02/05/2006.

He has presented his project work in our school titled as : **POST PARTUM CUP : A NOVEL WAY TO ESTIMATE THE OBSTETRIC BLOOD LOSS ACCURATELY AND SAVE MATERNAL LIFE IN GOLDEN HOUR.**

The Project work has been done by the student himself. He was motivated to peruse this innovative work by his science – mentor Mrs. Sangeeta Khanna and myself. The present novel idea is the “Brain Child” of his previous project – “Turn our red the green” for Homibhabha Balvaideyanik competition 2017-18, which was honoured with Gold Medal.

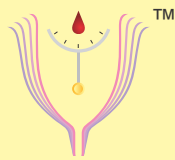
He holds the IP (intellectual property rights) for his product. The novel invention is being used in clinical practice by obstetricians. It has been widely appreciated by the experts in the medical field, as **life saving and eco- friendly.**



Signature
17/03/2020
Principal
Vivekanand Pratishthan's
Kashinath Palod Public School, Jalgaon (M.S.)

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POST PARTUM CUP

PAPER TITLE : **POST PARTUM CUP**

**A NOVEL WAY TO ESTIMATE THE OBSTETRIC
BLOOD LOSS ACCURATELY AND SAVE
MATERNAL LIFE IN GOLDEN HOUR.**

PAPER AUTHOR : ARCHIT RAHUL PATIL

KASHINATH PALOD PUBLIC SCHOOL, JALGAON

ABSTRACT

CURRENT SCENARIO

PPH is major cause of maternal death (25-55%).

Globally one mother dies every 4 minute due to PPH.

We are still living in the Shadow of Tajmahal which reminds us the grim tragedy of death of Mughal Empress Mumtaz who died during 14th child birth. We also lost beloved Actress Smita Patil due to Complications after childbirth in modern era at the premier medical institute : BREACH CANDY HOSPITAL, MUMBAI where all state of the art facilities were available.

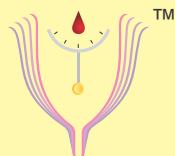
STATEMENT OF NECESSITY

PPH is very unpredictable and every parturient woman is at risk of having PPH.

*PPH-Post Partum Haemorrhage is all about -**P**redict **P**repare and **H**andle.*

*The accurate estimation of blood loss is most important key factor before the clinical signs appear and it is often under estimated in present clinical practice .Because of delay in diagnosis GOLDEN HOUR is missed and patient lands up in a **near miss** morbid condition in a state of shock which can cause death of mother.*

Timely resuscitation is must for better survival.



POST PARTUM CUP

NOVEL INVENTION: REMEDIAL MEASURES

*I have come up with the **unique, novel** idea & designed postpartum **cup** for **realtime estimation of obstetric blood loss**.*

*Post Partum Cup is the brain child of previous project titled-“Turn our **red** the **green**” for Homibhabha Balvaidyanik competition 2017-18, honored with gold medal.*

The idea struck to my mind; as my parents are obstetrician and gynecologist and anesthesiologist, they get distress calls for PPH at the odd hours and later I used to hear their discussions about the delay in diagnosis and the MARATHON efforts all medicos need to put to save the life of mother.

With the knowledge of menstrual cup- a collecting system of blood during menses ;which my volunteers are using comfortably; I thought of designing a novel postpartum cup which can be easily fitted into the vaginal tract and the blood loss estimation will be accurate & real time.

Implementation and conclusion

We used the postpartum cup in the clinical practice in four maternity homes under the guidance of obstetricians and found that,

*it is **life saving**, practical, convenient, healthier, economical, comfortable, easy to use and **ecofriendly**.*

*Postpartum cup is: **Cash free, rash free, trash free** same as its predecessor : menstrual cup*

*The **patients** are extremely happy, comfortable and readily acceptable to the new idea as it minimized the handling and excluded the feeling of wetness of postpartum blood loss.*

*The **nursing staff** found it user friendly to handle the postpartum cup, reduced the workload of cleaning, it eliminated the use of sanitary pads and subsequent sanitary waste.*

*The **obstetrician and gynecologists** welcomed the idea, they confessed that first time in their clinical practice, and they could see the real time estimation of post partum blood loss. They **got early** alarm of postpartum hemorrhage and manage the patients with timely measures and saved maternal lives .Now **obstetricians** are over demanding the samples of postpartum cup .The obstetricians have incorporated this invention in their routine practice and recommending to other doctors to use it. There is more demand for this postpartum cup from medical fraternity*

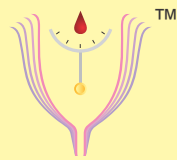
*This idea is to eliminate the **under diagnosis** and **loss of golden hour**. So timely measures are taken by the treating obstetricians and team to curb the maternal mortality.*

*This **life saving innovation** has advantage of eliminating the sanitary biomedical waste and is eco friendly.*

Commercial impact

*This novel idea is applied for **patent** as no such simple yet indispensable medical tool is available in medical literature. It also has great **commercial potential**.*

It needs to be presented at the global platform.



POST PARTUM CUP

I. INTRODUCTION

MOHD. FATAHLLA, PRSIDENT OF FIGO 1997:

“WOMEN ARE NOT DYING BECAUSE OF A DISEASE WE CAN NOT TREAT. THEY ARE DYING BECAUSE SOCIETIES HAVE YET TO MAKE THE DECISIONS THAT THEIR LIVES ARE WORTH SAVING”. Every parturient woman is at risk of having PPH. To curb the maternal mortality, early diagnosis even before appearance of clinical signs and symptoms is necessary. So golden hour is not lost.

“It is extremely difficult to make complex things simple whereas it is easy to make simple things complex” Steve Jobs

The passage of time likely to increase the complexity of any given case, because continuous bleeding if not appropriately and adequately controlled on a timely basis, invariably leads to coagulopathy.

PPH is Easy to miss

Physicians underestimate blood loss by 50%. So estimate blood loss inaccurately.

Slow steady bleeding can be fatal. Evaluate all bleeding, including slow bleeds.

Abdominal or pelvic bleeding can be hidden. If mother develops hypotension, tachycardia or pain, rule out intra-abdominal blood loss.

Estimation of accurate blood loss in the present clinical scenario is very difficult and impossible with the presently practiced and suggested method like....

A **pictorial** reference guide to aid visual estimation of blood loss at obstetric hemorrhage

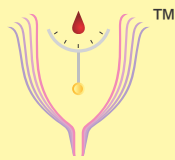
Blood drained into an fixed container for measurement

BRASSS-V DRAPE

To overcome this limitation and for real-time estimation of obstetric blood loss, with knowledge of menstrual cup, our novel idea, medical inputs and technical help, I designed postpartum cup.

I discussed with obstetricians and implemented its use in clinical practice. I observed various parameters, collected feedbacks from patients, nursing staff and obstetricians.

I concluded that this postpartum cup is readily acceptable by patients ,it is user friendly to nursing staff, effective in alarming about PPH and estimating the postpartum blood loss in real-time like never before. It also eliminated the sanitary waste to great extend.



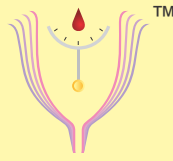
POST PARTUM CUP

II. DETAILED EXPLANATION

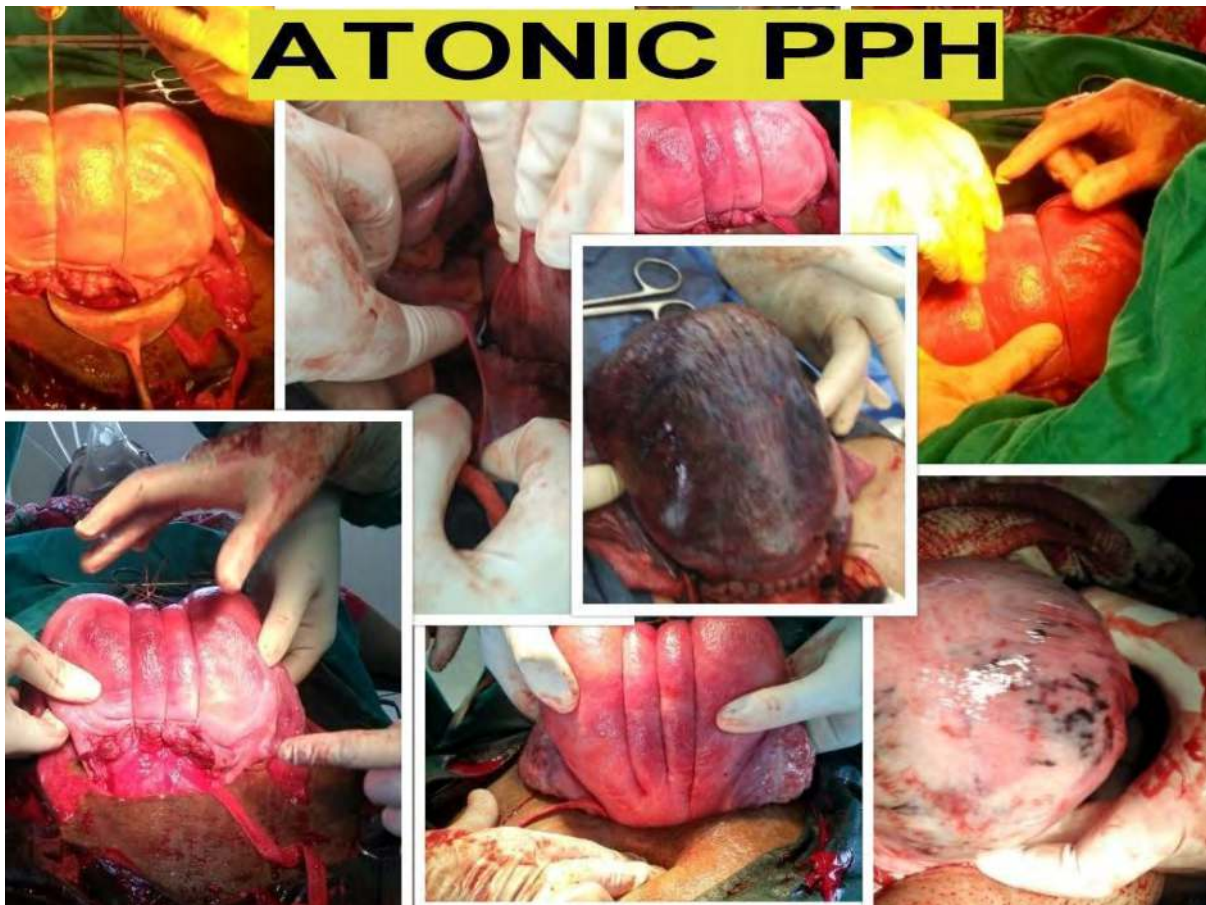
JOURNEY OF NOVEL INNOVATION – As my father is obstetrician and mother is anesthesiologist, they get distress call for help in cases of postpartum hemorrhage at odd hours and later I hear their discussion how marathon efforts were needed to put by all medicos to save the maternal life {like operative interventions- compression sutures, various balloon tamponade, stepwise de-vascularisation : even internal iliac artery ligation ,sometimes obstetric hysterectomy ; when all measures failed. (I used see photos and videos of these procedures)} . The delay or under diagnosis was responsible for landing the mothers in near- miss morbid condition which might cause maternal death. I thought ,what can be done to help obstetricians to eliminate the delay and can do real time estimation of post partum bleeding? The new method should be user friendly and acceptable to patient, nursing staff and obstetrician, which is not possible with prevalent measures available in clinical practice.

I did project titled-“**Turn our red the green**” for Homibhabha Balvaidyanik competition 2017-18 honored with **gold medal**. I found that menstrual cup made up of medical grade silicone is a **cash free ,rash free and trash free**, menstrual blood collection product is being used and is very good menstrual hygiene option.

During my project, I found excellent acceptance of menstrual cup by volunteers including my mother and other three gynecologists. **The idea struck to my mind, if this menstrual cup is modified to fit the drainage tube and connected to collection bag ,like a urine bag , it can be used in postpartum patient and will do real-time estimation of postpartum bleeding** .So I made a working model with the spare menstrual cup of my mother.



POST PARTUM CUP



Turn our **Red** the **Green**



Miserable Ground Reality @ Jalgaon



Interactive Workshops



Interactive Session at Nursing Students



Turn our **Red** the **Green**

School of Environmental & Earth Sciences (North Maharashtra University, Jalgaon)



Print Media

जळगावातील 'लिटल पॅडमॅन' अर्चित पार्टील

नॅपकीनच्या धुरामुळे कॅन्सर, वंध्यत्व, अंडाशयाचे आजार होण्याची शक्यता

जळगाव : सध्या सर्वत्र 'पॅडमॅन' या चित्रपटाची सर्वत्र चर्चा असून मासिक पाळी या विषयावर आजही लोक खुली चर्चा करीत नाहीत. त्यामुळे अशा विविध विषयांना हात घालण्याचा प्रयत्न पॅडमॅन मधून करण्यात येत आहे. याच अनुषंगाने जळगावच्या एका सहाय्येच्या विद्यार्थ्याने सॅनिटरी नॅपकीनच्या वाढत्या वापरापुढे होणारे आजार व प्रदूषणाच्या आज्ञे पाळण्यासाठी एक प्रोजेक्ट तयार केला आहे. या प्रोजेक्टमधून त्यावर उपाययोजनाही सांगितल्या आहेत.

मासिक पाळीसारखा महत्वाचा विषयवर आजही आपल्याकडे गुलती पाळली नाही. मासिक पाळीदरम्यान स्त्रियांचे अस्वस्थ, स्वच्छता वसाखला विषयावर बोललेलेही नवत नाही. नेमका याच विषयावर अखंड कुमार या 'पॅडमॅन' भाष्य करणार आहे. मात्र, यापेक्षाही दोन पाऊले पुढे जाऊन जळगावातील कॉमिनाथ पलीड या शाळेतील सहाय्येच्या विद्यार्थी अर्चित पार्टील याने एक प्रोजेक्ट तयार केला आहे. सॅनिटरी नॅपकीनचा टिमेंसीटस वाढता वार अर्ध तासातून निर्माण होणारा कचरा व प्रदूषण टाळण्यासाठी करवयाच्या उपाययोजना या प्रोजेक्टमधून दर्शविल्या आहेत.

असा निवडला विषय
अर्चित हा सहाय्येतील विद्यार्थ्याने डॉ. राहुल पार्टील व भुलनल डॉ. अर्चना पार्टील यांचा

एक जण निर्माण करत १२० ते १५० किलो कचरा
मासिक पाळीदरम्यान मुली व महिलांकडून सॅनिटरी नॅपकीनचा वापर मोठ्या प्रमाणावर होत आहे. पुर्वी वापरण्यात येणाऱ्या कापडाचा वापर हा कमी झाला असून प्रतिठेचा विषय हा बनला आहे. त्यामुळे एका वर्षासाठी एक महिला मासिक पाळीदरम्यान १४४ सॅनिटरी नॅपकीन वापरते असे एका सर्वेक्षण समोर आले आहे. म्हणजेच १३ ते ५० व्यापरीत एक महिला ५,३२८ एवढे सॅनिटरी नॅपकीन वापरते. परिणामी सॅनिटरी नॅपकीनचा एक महिला आपल्या आयुष्यात १२० ते १५० किलो कचरा निर्माण करते, असेही सर्वेक्षणानुसार समोर आले आहे.

अशा आहेत उपाययोजना
सॅनिटरी नॅपकीनवर पेट साईडचा वापर केलेला असतो. त्यामुळे हे जळोना जाऊन हानिकारक वायू बाहेर पडतात व कॅन्सर, अंडाशयाचे आजार उद्भवण्याची शक्यता असते. तसेच अशा नॅपकीनाला रिसावकलही करता येत नाही. किंवा नष्ट करण्यासाठी येण्या नाही. त्यामुळे हे प्रदूषण रोखण्यासाठी काय उपाययोजना करता येतील याबाबत माहिती घेतली असत माहवारी कथ व जुने लोक वापरत असलेले कपडे हाच पर्याय वापर असल्याचे समोर आले. माहवारी कप हा ऑनलाईन ऑर्डर केल्यास दोन दिवसात घरपोच मिळतो. एकदा पेटल्यावर तो द्या र्व उपयोगात येतो. याचा वापरबाबत इंटरनेटवर यु ट्यूबवर भरपूर माहिती उपलब्ध आहे. या कार्यावळी खोली चर्चा होत नसल्याने ते काही मौजगार्या लोकांपर्यंत पोहचले आहे. तसेच पुर्वीप्रमाणे कपडे वापरण्याने देखील योग्य आहे. हे कपडे घुऊन परत वापरात येतात. त्यामुळेही प्रदूषण टाळता येईल. असे डॉक्टरांचे म्हणणे आहे.

Electronic Media

जळगावचा 'पॅड मॅन'

जळगावचा 'पॅड मॅन'

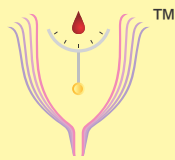
जळगावचा 'पॅड मॅन'

जळगावचा 'पॅड मॅन'



https://www.youtube.com/watch?v=dzwh7ou2a_E





POST PARTUM CUP



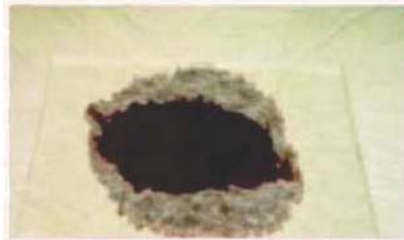
soiled sanitary towel
30ml



saturated small swab (10x10cm)
60 ml



saturated sanitary towel
100 ml



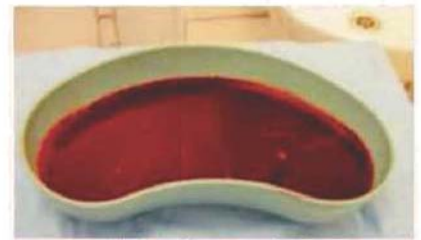
incontinence pad
250 ml



saturated swab (45x45cm)
350 ml



floor spill (100cm diameter)
500 ml



full kidney dish
500 ml



blood spilling on bed
1000 ml



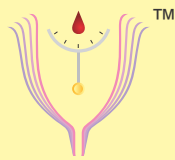
blood spilling to floor
2000 ml

Brass-V Drape



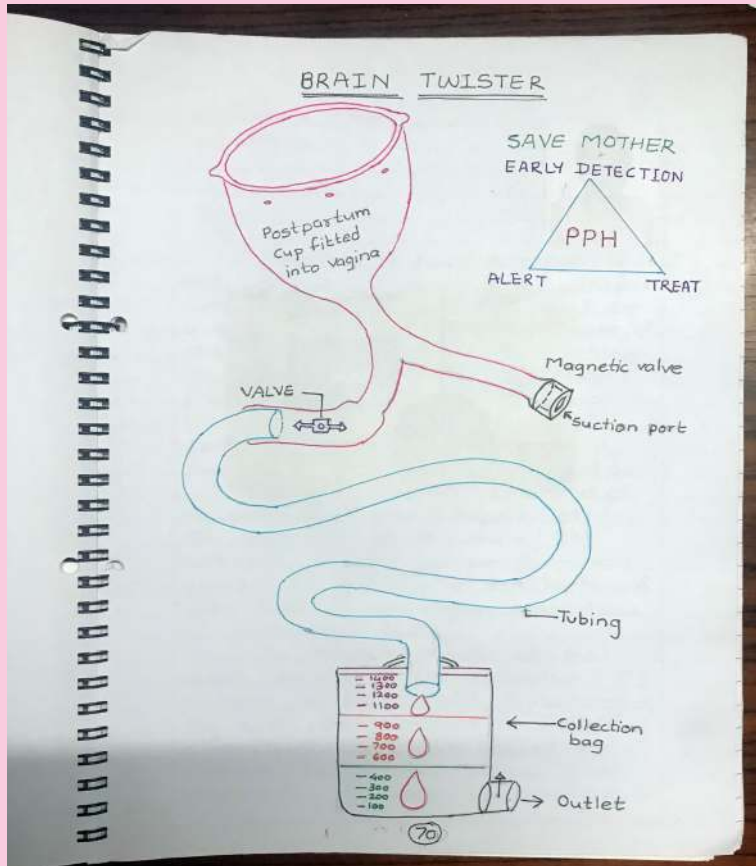
Kelly's Pad



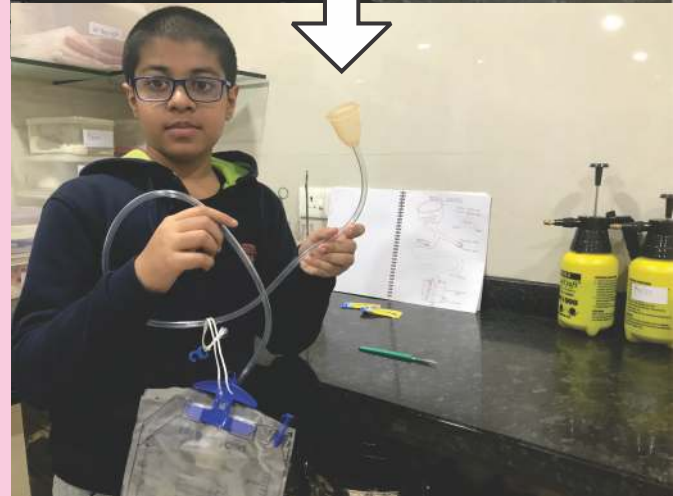
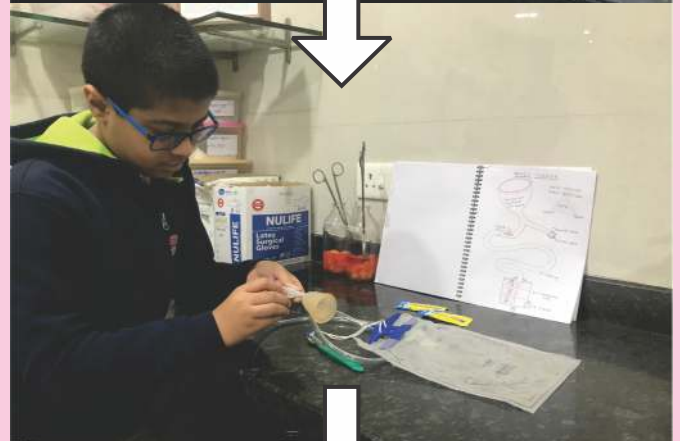


POST PARTUM CUP

First the idea was put on the paper.



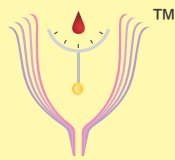
Preparing primitive working model



Presenting the Primitive working model & Idea of PPH Cup to Gynecologist.

Primitive working model project presented at KPPS, Jalgaon at School Level.





POST PARTUM CUP

I presented this idea to four obstetricians. I discussed the practical feasibility and their concerns, also listened to suggested modification in the final postpartum cup which can be helpful in clinical practice.

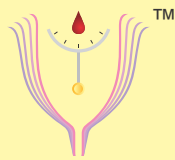
The **concerns** and **suggestions** in temporary working model were –

- 1) Need for modification in size-little bigger than menstrual cup considering the postpartum roominess of birth canal in parturient women.
- 2) There can be blockage due to blood clot so need for the suction port with larger inner diameter.
- 3) There should be patient comfort and easily acceptable.
- 4) There should not be any leakage by the side of post partum cup or suction port
- 5) The cup should remain in place and should not come out during shifting of patient.

With these concerns, we finalized the dimensions of the cup and drainage tube; after discussion and consensus among obstetrician.

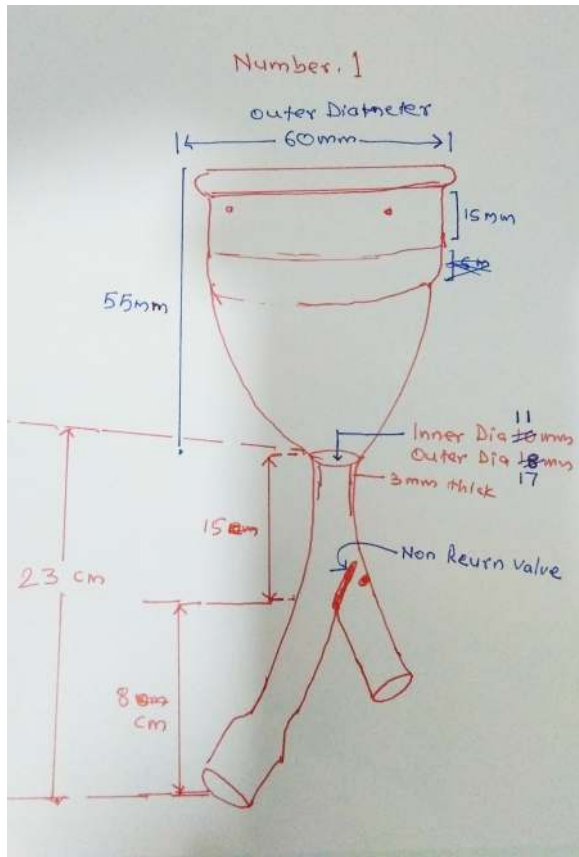
Then with this much idea, working model, final layout of postpartum cup and its dimensions, I consulted technical persons in MIDC, Jalgaon to get the samples ready. For that we along with technical guidance designed the mould and its drawings. I got three sample pieces made up of medical grade silicone. Doctors checked its feasibility, suggested shortcomings to modify the mould if needed, before finalizing the mould. As the non return valve in suction port was not feasible in sample piece so we deleted it in the final mould design and instead used the cap of urine bag as a temporary obturate.

Final mould was ready after modification .We manufactured 5 pieces of postpartum cup made up of medical grade silicone which is readily autoclavable for reusing it. The postpartum cup was introduced to the four obstetricians and they willingly started using it in clinical practice, we studied the parameters related to patients, nursing staff and obstetricians.

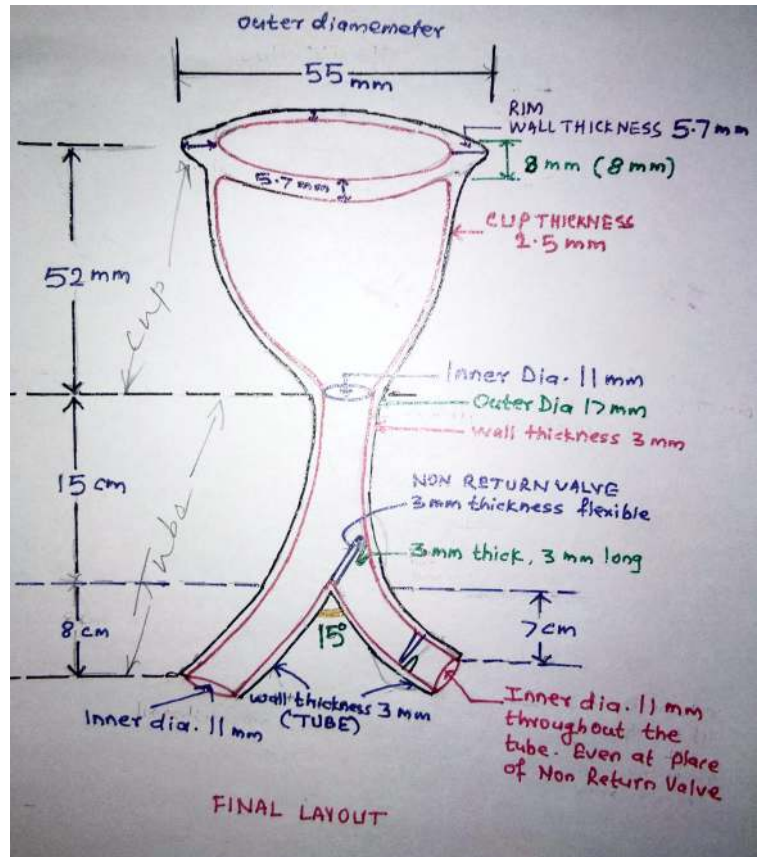


POST PARTUM CUP

Developing 2D Layout of PPH Cup



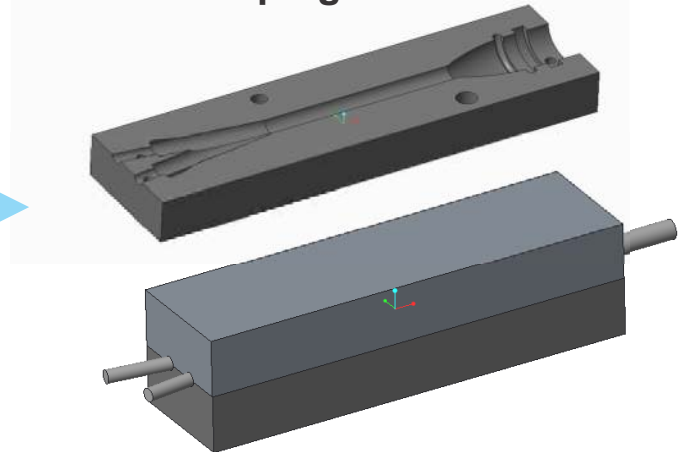
Final Layout and fixing dimensions for production of PPH Cup



Developing 3D Model



Developing Mould

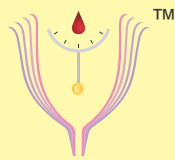


Actual Mould



PPH Cup ready for clinical use.

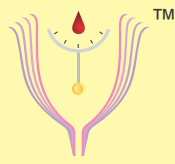




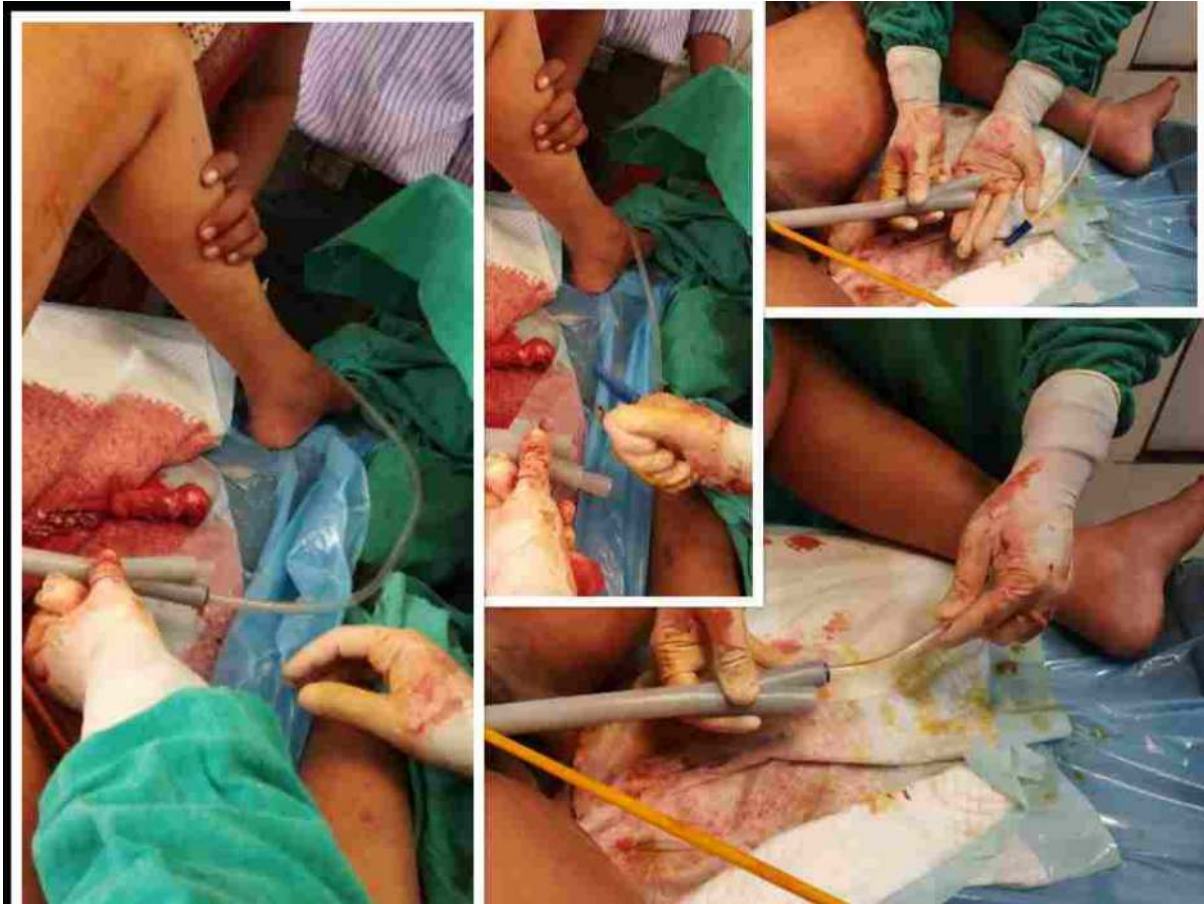
POST PARTUM CUP



**POSTPARTUM CUP WITH SUCTION TUBE, USED DURING THE PROJECT
AT MATERNITY HOSPITALS**

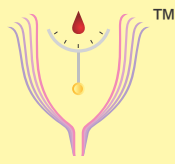


POST PARTUM CUP



ROUTINE USE OF POSTPARTUM CUP IN
CLINICAL PRACTICE

CLINICAL USE OF POST PARTUM CUP



POST PARTUM CUP



Clinical use of PPH Cup.

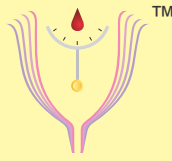


Bed side monitoring of post delivery blood loss in collection bag.



Realtime & Accurate measurement of post delivery blood loss with the help of PPH Cup.





POST PARTUM CUP

Doctor's Feedback

कृष्णा हॉस्पिटल प्रसूतीगृह आणि बालरोगालय

डॉ. संजय बाविसकर
M.D. (Mumbai), D.N.B. D.C.H.,
F.C.P.S. M.N.A.M.S.
नवजात शिशु व बालरोग तज्ज्ञ

डॉ. सी. अनिता बाविसकर
M.S. (Ob. & Gyn. Mumbai)
D.G.O. (Mumbai)
सर्वीस व प्रसूतिगर्भ तज्ज्ञ

स्वयंजामिया रोड, प्रतापनगर, टुंगीक गार्डन समोर, जळगाव. फोन : 0257-2223320, 2223321

नांव : _____ दि. / / 201

पता : _____

Appreciation letter...

I myself Dr. Anita Sanjay Bawiskar working as consultant obstetrician and gynaecologist at Krishna hospital, Jalgaon wish to share my opinion regarding use of post partum cup- "ARCHIT PPH CUP"

I am first user of this Post partum cup & using it since last more than 2 years routinely.

It is of great use to measure the blood loss after delivery. I can take immediate action for post partum haemorrhage. Prompt treatment is life saving. It also helps in high risk patients for PPH e.g., Over-distended uterus: polyhydramnios, twin pregnancy, PIH, DIC, etc.

For obese patients to change frequent pads after having heavy bleeding is very difficult and requires manpower. Here with the help of suction machine we can remove blood clots easily and thus physical handling of patient, which is very painful can be definitely minimised. It also saves patients money as no requirement of changing pad for first 24 hours. No staining of bed sheet and patients cloths too.

I really found this innovative idea of postpartum cup for patients user-friendly. We can save life of patients suffering from PPH, which is one of the major causes of maternal mortality in our country. Thus, we can help to reduce maternal mortality rate and achieve our goal for maternal health

Date-07/07/2020


ANITA BAWISKAR
Obstetrician & Gynaecologist
M.S. D.G.O.
Reg.No.2002/03/1129
फैर तपासणी : _____

रत येतांना हा कागद सोबत आणणे आवश्यक आहे.

डॉ. जितेंद्र मोरे
M.B., D.G.O.
स्त्रीरोग व प्रसूतीशास्त्र तज्ज्ञ
Reg.No. 2001/08/2830

ऑकार नगर, पांडे डेअरी चौक, जळगाव. फोन : 0257 - 2233899

पेसंटचे नांव : _____



दिनांक : _____



29 February 2020

I am writing this appreciation letter as an obstetrician & user of POST PARTUM CUP.

I am witness to the genesis of idea of designing of Post Partum Cup by modifying the menstrual cup to its present version of ARCHIT PPH CUP.

It gives exact measurement of post delivery blood loss, which was never possible before. It is a boon for every obstetrician. It is lifesaving and eco friendly as well.

I am one of the first users of this ARCHIT PPH CUP in clinical practice since November 2018. I have used in 215 patient. It is very user friendly, comfortable for patients. It gives piece of mind when it comes to monitoring the post delivery blood loss. It has helped me in promptly managing 8 cases of PPH, in golden hour due to early recognition of PPH. I could save those precious life of mothers without morbidity.

I congratulate Archit for providing this major breakthrough which everyone was looking for. It will be major milestone in history of PPH.

I recommend every obstetrician should use this tool in their clinical practice.


Dr. Jitendra More
Moraya Maternity Hospital
Morya Hospital, Jalgaon
Dr. JITENDRA B. KOLI
M.B.B.S., D.G.O.
Reg.No.2001/08/2830

फैर तपासणी : _____

तपासणी वेळ : सकाळी १० ते २, संध्या. ५ ते ७ (रविवार - फक्त अपॉइंटमेंट घेऊन यावे)

Patient's Feedback

पेसंटचे नाव :- अश्विनी चेतन कोपले
पता :- चंद्रप्रभा कॉलनी, प्लॉट नं. १९ जळगाव

Mob :- _____

डिलिव्हरीची तारीख :- 26/12/18

माझी डिलिव्हरी नंतर मी Post partum cup वापर केला Post partum cup हा उत्तम पर्याय आहे कारण माझी पहिली डिलिव्हरी ही दुसऱ्या हॉस्पिटलमध्ये झाली तिथे मला प्रकारचा Post partum cup वापरला नाहीता, ज्या Post partum cup मुळे वारंवार पॅड बदलावे लागले नाहीत. आणि जे खोणारे बिल्लींग योग्य वेळी सफाई करून काढता आले. माझे मला कुठल्याही प्रकारचा त्रास झाला नाही. भिचमिच सुरक्षा वाटले नाही. Post partum cup हा माझ्यासाठी परित्या डिलिव्हरी मध्ये खोणाला त्रास पेक्षा चांगला उपयोगी पडला.

सुधी
अश्विनी चेतन कोपले

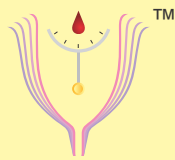
दिवाली हर्षल चाळुंवे
वय :- 28
राधाकृष्ण नगर जळगाव
ता. जि. जळगाव
Mob No :- 9822375100
Delivery Date :- 13/11/2018

PostPartum cup हा माझ्या Delivery मध्ये दिवशी मला use करूयात आला. त्यामुळे Pad घेणे हा त्रास तर नव्हता पण मला त्यामुळे फ्रेश वारले व माझे blood किती गेले हे पण कळले. तपकील परत परत change करणे हा त्रास तर नव्हताना. Postpartum Cup हा अगदी उत्तम पर्याय आहे.

Postpartum Cup हा Best option तर आहेच पण हा एक great experience आहे.

अश्विनी विशवाय

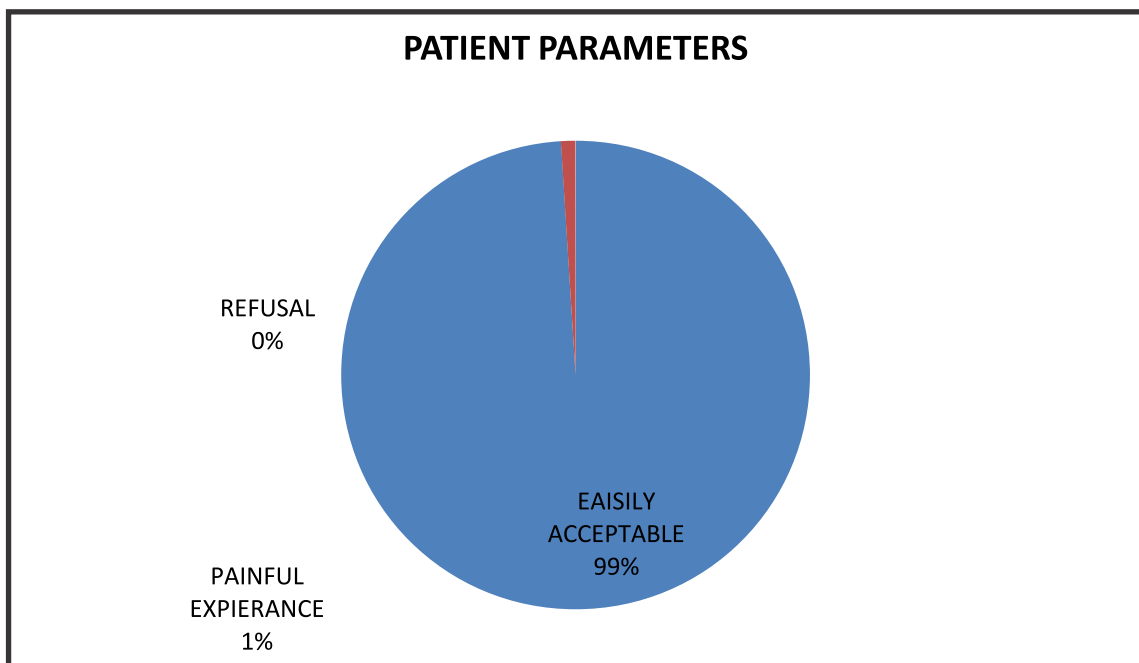
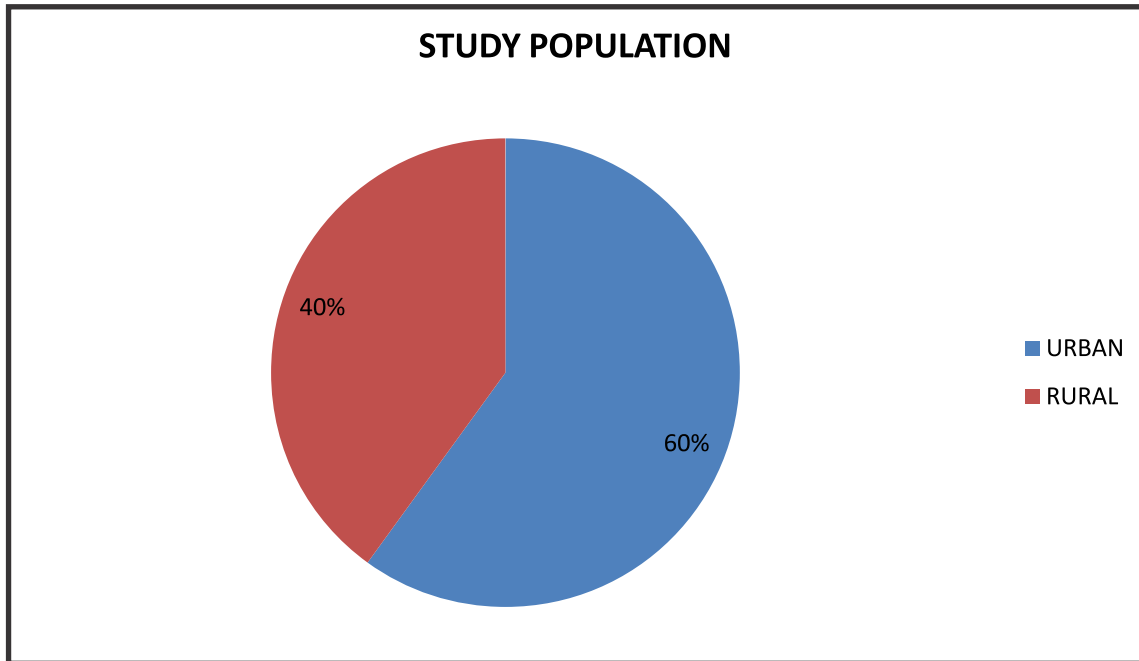
26/12/2019

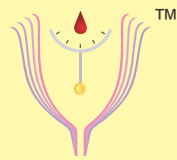


POST PARTUM CUP

III. OBSERVATIONS

PATIENT RELATED PARAMETERS

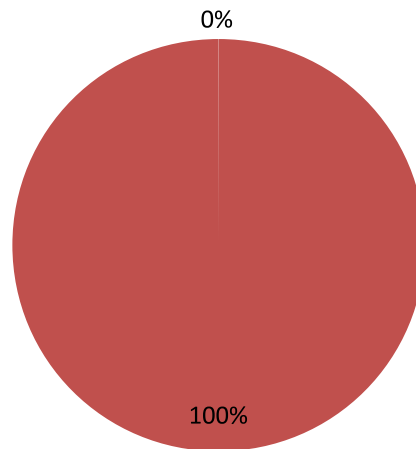




POST PARTUM CUP

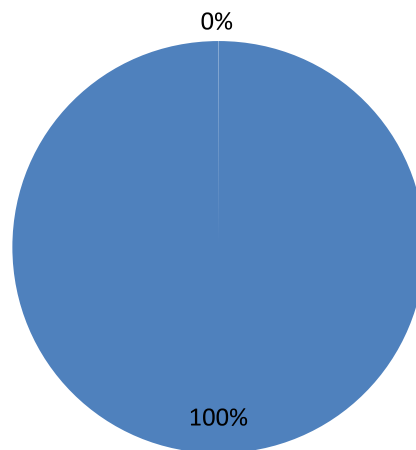
FEELING OF WETNESS TO PATIENT

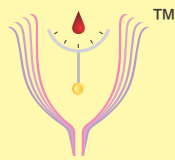
■ YES ■ no



SATISFACTION ABOUT NEW METHOD

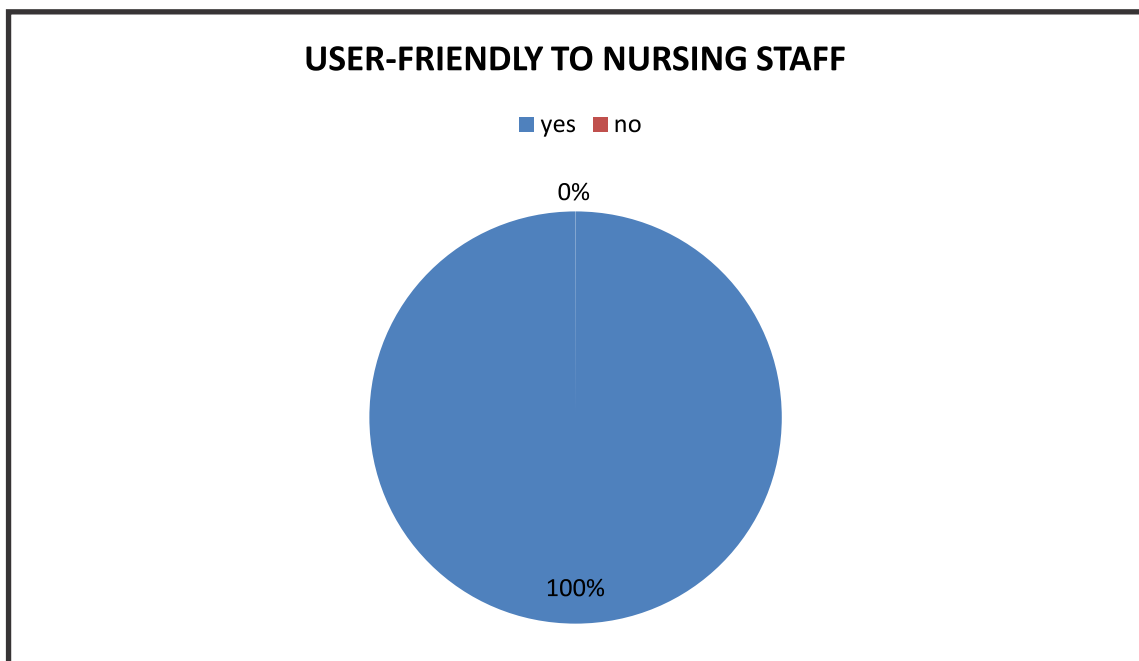
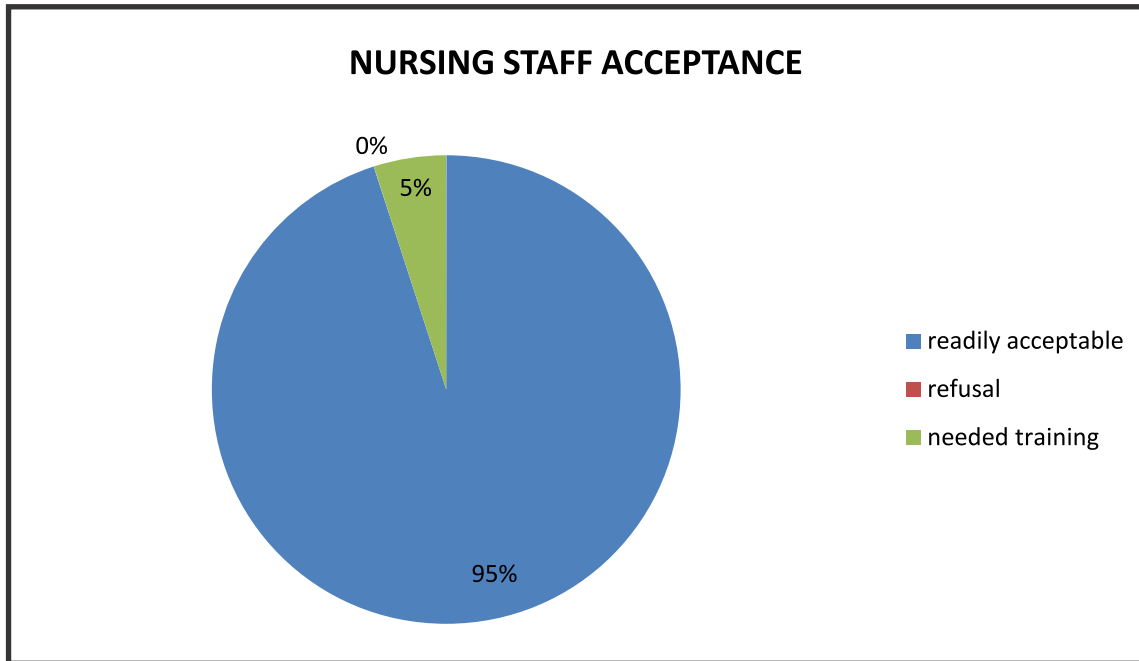
■ HAPPY ■ UNHAPPY

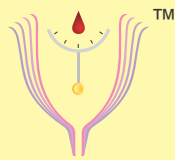




POST PARTUM CUP

NURSING STAFF RELATED PARAMETERS

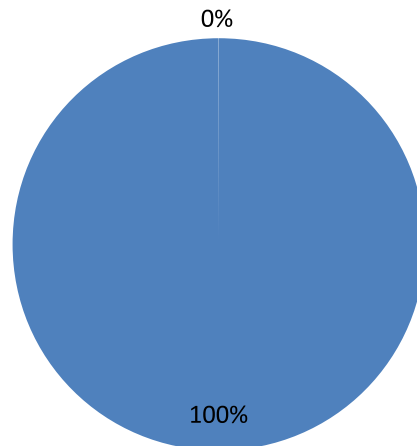




POST PARTUM CUP

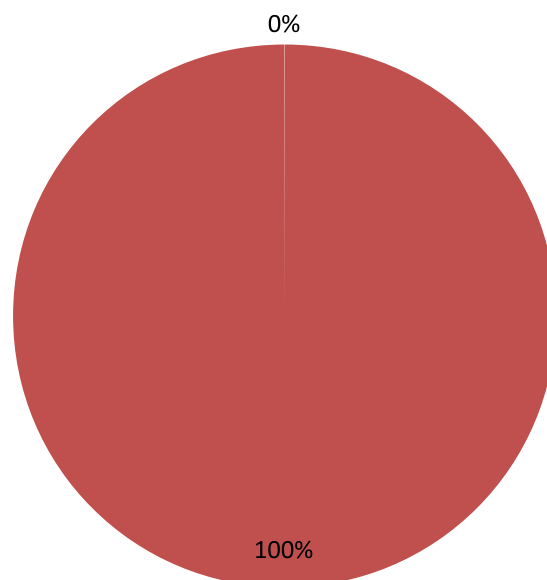
LEARNING CURVE OF NURSING STAFF IN DOING SUCTION

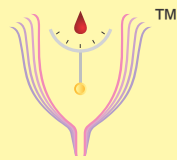
■ eaisy ■ difficult



MESSINESS IN HANDLING THE PATIENT/POSTPARTUM CUP BY NURSING STAFF

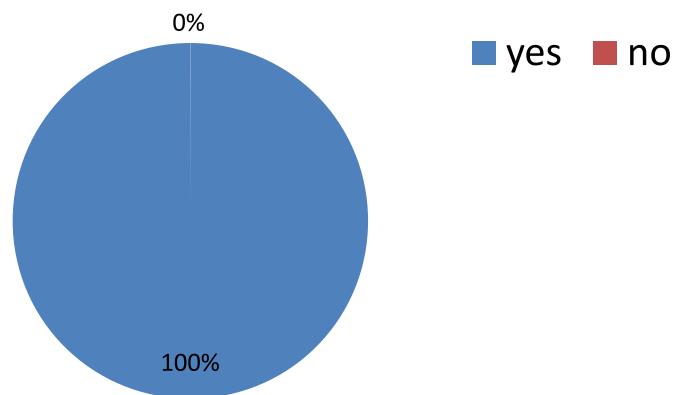
■ yes ■ no



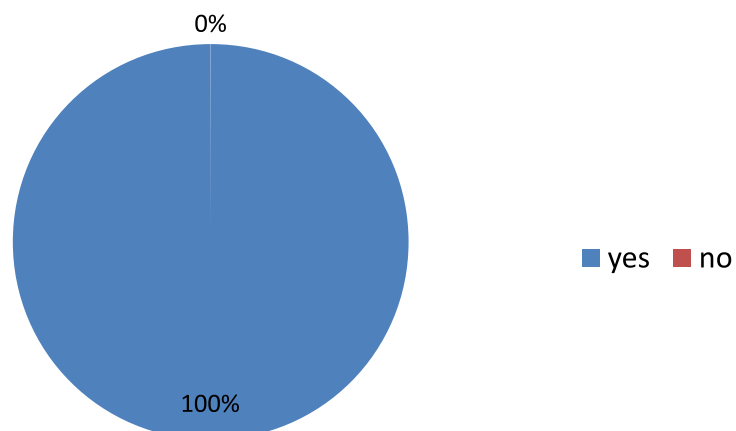


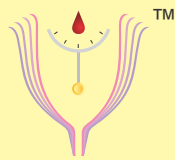
POST PARTUM CUP

TIMELY EARLY ALARM ABOUT EXCESSIVE POST PARTUM BLEEDING



REDUCTION USE OF SANITARY PADS AND PRODUCTION OF SANITARY WASTE



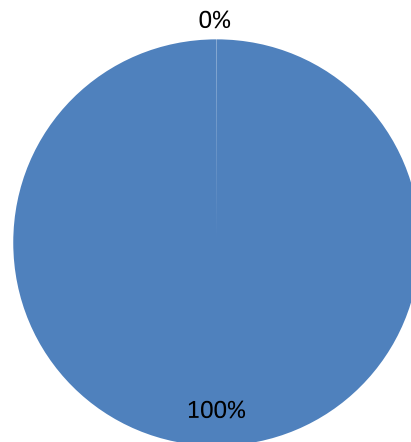


POST PARTUM CUP

OBSTETRICIAN RELATED PARAMETERS

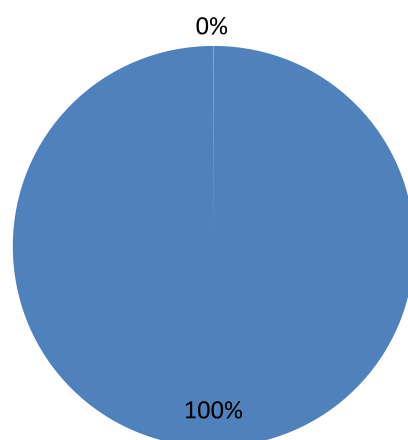
ACCEPTANCE BY OBSTETRICIANS TO POSTPARTUM CUP

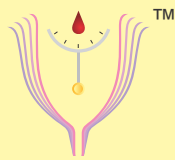
■ readily accptable ■ refusal



INSERTION/REMOVAL TECHNIQUE BY OBSTETRICIANS

■ easiy ■ difficult

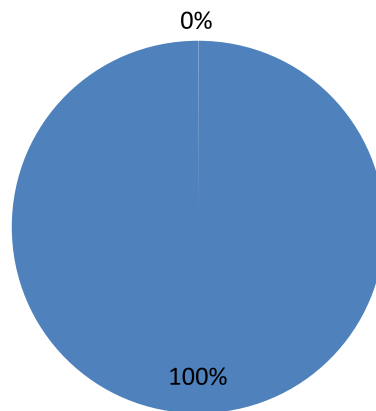




POST PARTUM CUP

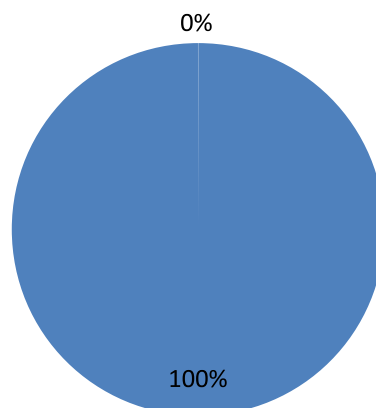
OBSTETRICIAN FOUND HELPFUL IN MANAGEMENT OF PPH DUE TO EARLY RECOGNITION

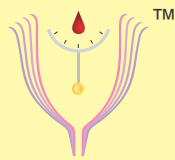
■ yes ■ no



DEMAND FOR MORE SUCH SAMPLES OF POSTPARTUM CUP

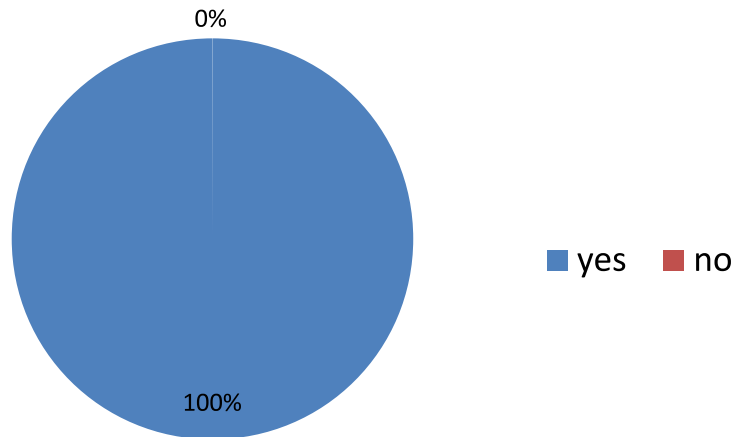
■ yes ■ no



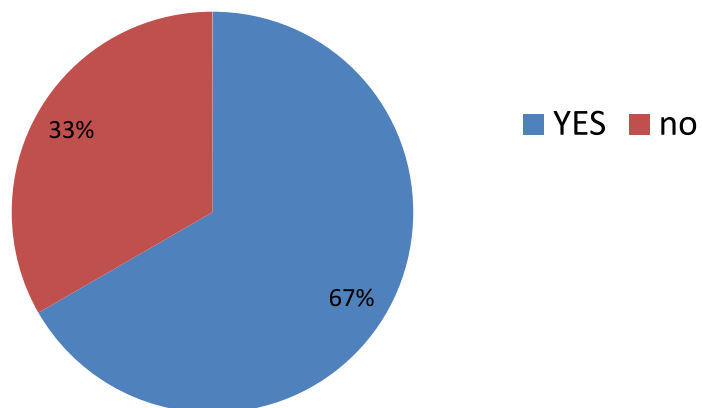


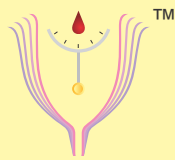
POST PARTUM CUP

WILL RECOMMEND TO OTHER OBSTETRICIANS



SUGGESTION ABOUT DIFFERENT SIZE

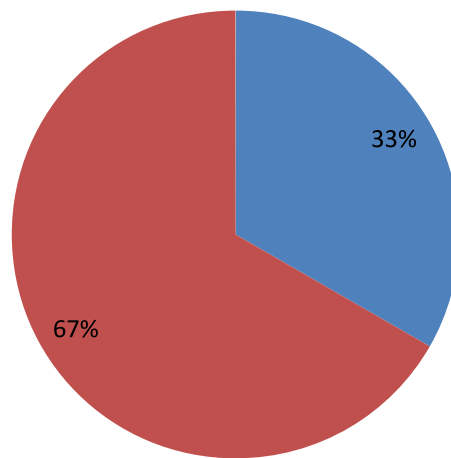




POST PARTUM CUP

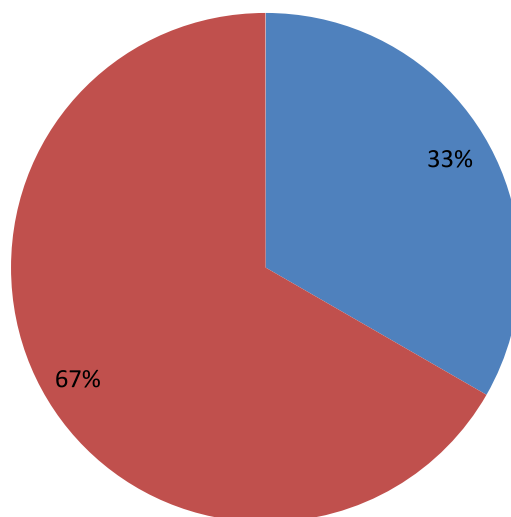
SUGGESTION ABOUT INCREASING THE INNER DIAMETER OF DRAINAGE SYSTEM

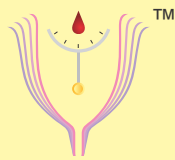
■ YES ■ NO



SUGGESTED MORE SNUGLY(TIGHT)FITTING OF URINE BAG TO DRAINAGE TUBE OF POSTPARTUM CUP

■ yes ■ no





POST PARTUM CUP

IV. CONCLUSION

After introduction of this novel idea of postpartum cup in the actual clinical practice from the observations, feedback of patients and obstetricians I concluded,

ADVANTAGES:

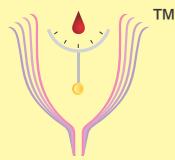
- 1) The postpartum cup was readily acceptable to patient, user friendly to nursing staff and obstetrician.
- 2) Patients were extremely happy after its use as there was total dry feel after delivery. Very less handling of patients while checking the obstetric blood loss.
- 3) The postpartum cup was safe and effective in real-time estimation of post partum bleeding which was not possible with prevalent methods.
- 4) By using Post partum cup the obstetricians were alarmed early about excessive post partum bleeding so patients were treated without losing golden hour.
- 5) There was reduction in use of sanitary pads so less sanitary waste was generated.
- 6) Obstetricians easily accepted this novel idea to be implemented in clinical practice, they found insertion and removal techniques very easy. The cup formed watertight seal with vaginal walls and there was absolutely no leakage of blood from the side of postpartum cup.
- 7) Through suction port, easy suction was possible; so no blockage of the system due to blood clots. Specially more useful in overweight patients as there was no handling of patient.
- 8) The obstetricians, who conducted the trial were happy using it and demanded more such postpartum cup. They also recommended use of postpartum cup to their colleagues.
- 9) **Silicone** is used in a variety of medical instruments and equipment which must be sterilized before use. Three main methods of sterilization can be considered: steam sterilization (**autoclave**), irradiation and ethylene oxide.

LIMITATIONS / SUGGESTIONS:

Although there were no limitations, however few suggestions such as,

- 1) Different sizes should be made available.
- 2) The inner diameter of the suction port should be such that 10 mm suction tube should negotiate easily the joint of drainage end and suction end of the postpartum cup.
- 3) The urine bag should be snugly (tight) fitting to drainage end of the cup to make it water tight.

Above mentioned suggestions were implemented by modifications in the designs of same mould.



POST PARTUM CUP

IMPORTANCE OF THE WORK:

PPH is the major killer of mothers during childbirth all over the world.

This idea is to eliminate the **under diagnosis** and **loss golden hour**. So timely measures are taken by the treating obstetricians and team to curb the maternal mortality. The postpartum cup works wonder in accurately estimating the postpartum obstetric blood loss.

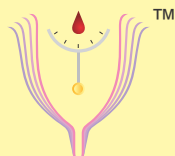
The things should start here and society should walk the talk to save maternal life. Every life is precious. This innovation will have impact to save many maternal lives if it is brought on the right platform and **reduce the maternal mortality rate as per policy of Government of India and WHO (world health organization)**.

This novel cup reduces the sanitary waste generated in hospitals to a great extent and so is **ECOFRIENDLY**

This novel idea is applied for **patent** as no such simple yet indispensable medical tool is available in medical literature. It also has great **commercial potential**.

It also needs to be presented on the global platform.

This project is selected for "NLEPC INSPIRE-MANAK AWARD-2020". Also awarded as "Youngest Innovator - MS SIGNATURE AWARD-2020"



POST PARTUM CUP

V. ACKNOWLEDGMENTS

The idea of developing scientific temperament has been brought up to me by my beloved principal Late Suhas Mule Sir who provided me constant concept learning questions and provided me platform to participate in interschool science competitions. I express my deep gratitude to him. I owe my deep gratitude to my mentor Mr. Amit Bhatiya, Principal, KPPS ,Jalgaon who took keen interest on our project work and guided us all along and allowing me to conduct workshop in our school.

I heartily thank Dr. Rajashree Sadulwad ; because of her suggestion about incorporating brain twister in my previous project. Taking inspiration from her I came up with novel idea of designing postpartum cup from menstrual cup. Special thanks to her for her guidance and suggestions during this project work. I also thank Dr. Anita Bawiskar and Dr. Jitendra Koli, Dr. Rahul Patil (my father) for using postpartum cup in clinical practice and sharing critical suggestions in designing the postpartum cup.

I would like to thank Mr. Sanjeev Patil-Suyash Industries and Mr.Vijay Patil–Vikram Industries MIDC Jalgaon for technical help in designing the drawings of mould. I am grateful to Mr. Aanand Matwandkar, Gayatri Rubber, Pune for manufacturing the postpartum cup .

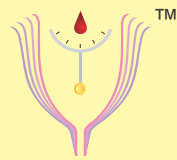
I would like to thank the **MINISTRY OF WOMEN AND CHILD DEVELOPMENT, GOVERNMENT OF INDIA** for giving me a golden opportunity to participate in this BAL SHAKTI PURSKAR competition and present this wonderful novel work.

I am thankful to and fortunate enough to get constant encouragement support from all staffs of hospital, patients and doctors who helped me in successfully completing my survey.

I would like to express my gratitude to my parents for giving me courage, motivating me and for providing necessary conditions to make the project successful one.

Special mention of my mother Dr. Archana Patil "champion of the women" for her love, support, encouragement and being my first volunteer of menstrual cup.

I am very grateful to all my dear friends, relatives & well wishers who encouraged & helped me to take up the task and complete it.



POST PARTUM CUP

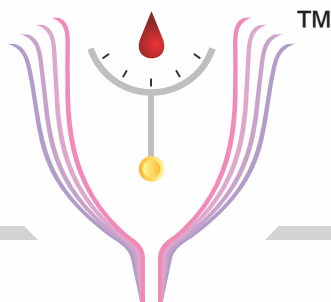
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2. WHO. Reducing the Global Burden: Postpartum Haemorrhage. *Making Pregnancy Safer*. 2007. [Full Text].
3. American College of Obstetricians and Gynaecologists. ACOG Practice Bulletin: Clinical Management Guidelines for Obstetrician-Gynaecologists Number 76, October 2006: postpartum hemorrhage. *Obstet Gynecol*. 2006 Oct. 108(4):1039-47. [Medline].
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6. Sentilhes L, Vayssi re C, Deneux-Tharaux C, et al. Postpartum hemorrhage: guidelines for clinical practice from the French College of Gynaecologists and Obstetricians (CNGOF): in collaboration with the French Society of Anesthesiology and Intensive Care (SFAR). *Eur J Obstet Gynecol Reprod Biol*. 2016 Mar. 198:12-21. [Medline].



Special Thanks

Project Guide	: Mrs. Sangita Khanna
Principal	: Mr. Amit Bhatia Sir, Kashinath Palod Public School, Jalgaon.
Valuable Inputs	: Dr. Rajeshree Sadulwad
Technical Guidance	: Mr. Sanjeev Patil
Mould Development	: Mr. Vijay Patil, Vikram Industries, Jalgaon
PPH Cup Production	: Mr. Anand Matwandkar, Shree Gayatri Rubber Industries, Pune
Clinical Application	: Dr. Anita Baviskar, Dr. Jitendra Koli
Parents	: Dr. Rahul Patil, Dr. Archana Patil
Platform	: Ministry of Women & Child Development Government of India
Volunteers, Patients & Hospital Nursing Staff.	
Animation	: Santosh Khopde, Wings Studio Jalgaon.
Graphics Design	: Amol Mahajan, Image Creation Jalgaon.



POST PARTUM CUP

Recommendation Letters



Recommended by		Designation & Institute
Dr. Poonam Varma Shivkumar		Professor & Head- Department of Obstetrics & Gynecology, MGIMS, Sevagram. National Resource Person for MH, MOHFW, GOI
Hon. Abhijit Raut, I.A.S.		Collector & District Magistrate, Jalgaon. (B.Tech. in Biotechnology from IIT)
Dr. Shyamkumar S. Sirsam		Medical Superintendent, Professor & HOD- Department of Obstetrics & Gynecology, Government Medical College & Hospital, Akola (MS).
Dr. Sharad Gokhale		Professor & Dean of Infrastructure, Planning & Management, Department of Civil Engineering, IIT Guwahati (Assam)
Dr. Shrinivas N Gadappa		Professor & HOD- Department of Gynecology & Obstetrics, Government Medical College, Aurangabad (MS).
Dr. Rajesh Rohilla		Professor, ECE Department, Delhi Technological University, Delhi-110042.



POST PARTUM CUP

DEPTMENT OF OBSTETRICS AND GYNAECOLOGY
MAHATMA GANDHI INSTITUTE OF MEDICAL SCIENCES SEVAGRAM
&
KASTURBA HOSPITAL, SEVAGRAM, WARDHA- 442102 MAHARASHTRA

Dr. Poonam Varma Shivkumar
Professor and Head
Department of Obstetrics and Gynaecology
Mgims, Sevagram



KASTURBA HOSPITAL SEVAGRAM WARDHA
Email:- poonamvarmask@yahoo.co.in
Institution: 07152284341 to 284355
Office :321, Resident:295, mob. 9881964571
Resident : 07152284127

I am pleased to endorse the application of ARCHIT RAHUL PATIL who has, in my opinion, accomplished great invention of a medical device: "post partum cup-The PPH Cup". He deserves the prestigious "**Pradhanmantri Bal Shakti Puskar**".

It has been a privilege to know this science and maths prodigy-Archit.

I came to know him on 20 February 2020; after he got selected for Inspire -Manak award, National level exhibition and completion 2020.

I came to know about his list of **innovative** and too many extraordinary **scholastic** achievements as well.

I went through all the details of his innovation, the "**Archit PPH Cup**".

I AM SO IMPRESSED THE WAY IT HAS BEEN DEVELOPED BY HIM AT THIS YOUNG AGE

I am sure he is a gifted child, super intelligent and extremely skilled. He has shown great problem solving aptitude.

Where the world is trying its different ways to ensure the estimation of blood loss, his PPH cup is such an innovation, which will revolutionize the PPH management and save number of women from mortality and morbidity.

I went through the videos, verified personally and realized that it is very simple technique by which it can be inserted. It can be used at lower resource settings too, where even ANMs can be trained to put it. It is very cost effective and easy to sterilize. It's a device which I feel should be brought to Global level.

It has been patented and is registered trademark. It is available locally and being used in clinical practice. Archit's this innovation must go to the national portal and it needs to be advertised nationally and internationally.

I will surely look into it and bring it to the notice of Ministry of Health & Family Welfare.

I would also like it to be seen by International people as it is a great device and very helpful in assessing the amount of blood loss and its replacement in the same proportion.

It will not only estimate but help us in taking decision for the management of dying PPH cases and prevent them from going into pulmonary oedema because of over enthusiastic transfusions.

In the past Archit has been awarded Gold Medal for Homibhabha Bal Vaidnyanik competition for his work on sanitary waste and menstrual hygiene options.

For THE PPH CUP invention, He is going to be felicitated for his incredible contribution to the society as a "**YOUNGEST INNOVATOR**" **MS SIGNATURE AWARD** an elite and prestigious award by group of Global NGOs and NRI's from 6 countries.

I wish Archit should be honoured for his life saving innovation which is economical and eco-friendly.

It will encourage him to bring out much more such innovations in the future.

I recommend him for the "**Bal Shakti Award**" (innovative category).

Dr Poonam Varma Shiv kumar ,
MBBS, MD OBGY, ACSHRH, SWEDEN,
Professor and Head, OBGY, MGIMS, Sevagram, Wardha ,MS
National resource person for MH, MOHFW,GOI

Date:- 14/07/2020

Professor & Head,
Obstetrics & Gynaecology
M.G.I.M.S., SEVAGRAM
WARDHA - 442102 (M. S.)



POST PARTUM CUP

Abhijit Raut I.A.S.

**Collector & District Magistrate,
Jalgaon**



D.O. Letter No. PA/E-Tapal/2020/56/127/2112

**Collector & District Magistrate Office,
Jilha Peth, Near Aakashwani,
Jalgaon - 425001.**

Telephone No. : 0257 - 2220400.

E-mail : collector.jalgaon@maharashtra.gov.in

Date : 18/07/2020

Recommendation Letter

Master Archit Rahul Patil presented his biomedical Invention-Post Partum Cup- Archit PPH Cup. I Appreciate the novelty, simplicity of use, its social contribution to the cause of reducing maternal deaths due to excessive bleeding after delivery by recognizing it early. In addition to it, it is reusable and eco-friendly. I can see a blend of academic excellence with innovative aptitude in this young boy. He has not only shown rare sensitivity towards the grave problems but also suggested the way to handle it. He has been appreciated and honoured with Inspire Award, "Youngest Innovator- MS SIGNATURE Award 2020, Gold Medal at Dr. Homibhabha Balvaidnyanik science talent search competition." In addition to this, He has won Gold Medals at International and National level in various Olympiad examinations. The list of his achievements is exhaustive.

We all know India is country of milleaneals. If all of them start thinking like Archit did, then I am sure India will become a superpower. He holds the intellectual property rights for this invention. Archit's aspiration to register a start-up is right towards "Make In India" Campaign.

To encourage this bright young innovator, I "highly recommend" him for the prestigious — **Bal Shakti Purskar (Innovative Category)**.

**Abhijit Raut
Collector &
District Magistrate Jalgaon.**



POST PARTUM CUP



Government Of Maharashtra

Government Medical College & Sarvopchar Rugnalay, Akola (MS)

Medical Superintendent

Phone No :- 0724-2421023, 2434401

Email :- msakolagmca@gmail.com

Fax No :- 0724-2437078

O.N.GMC&H/ MS / 238 / 2020

Date :- 15 / 07 / 2020

Recommendation Letter

I feel honoured to recommend **Master. Archit Rahul Patil** for the prestigious award: **Pradhanmantri Bal Shakti Purskar.**

I came to know about his novel innovation: **Post-Partum Cup- Archit PPH Cup** when he was selected for **NELPC Inspire Manak Award** at Amravati, on 8 February 2020. I read about him in news paper. I am **Medical superintendent and professor & Head, the Department of Obstetrics and Gynecology, Government Medical College, Akola (Maharashtra).**

To know more about Archit, I approached his parents. I am impressed by his exceptional achievements in science and maths since early childhood. He has participated in many science exhibitions. He has participated in prestigious mathematic camps: RAM-TP 2019 (raising a mathematician –training program) and MATHS.COM

Archit is a genius. He has long list of academic achievements at national and international level to his credit. He has secured Gold medals at various science and maths Olympiads, like TIMO, SEAMO, AIMO, ISTSE, Mathex, Ignited mind lab. He received **Gold Medal at Homi Bhabha Bal Vaidnayanik** competition. He was Maharashtra state scholarship – state topper. These are few mention out of the exhaustive list of his achievements.

His latest novel innovation ARCHIT PPH CUP is one more feather added to his cap. He is a **“YOUNGEST INNOVATOR – MS SIGNATURE AWARDS 2020”.**

His innovation was **highly rated by the ASME, ISHOW** organizer in a special letter.

This innovation is notable for the very obvious reasons. It is very simple, easy to use cost effective and eco-friendly. It measures the post-delivery blood loss accurately and in teal –time.

The excessive blood loss (post-partum Haemorrhage) is the leading cause of maternal deaths. The main reason for such death is delay in recognizing it.

This device is one stop solution to this problem. This life saving device can be very useful at public hospitals. I have procured the 6 devices from Archit and using it at Government Medical College, Akola since March 2020

The results are amazing! Soon I will publish the study in national and international journal.

This device will change the practice of assessment of post-delivery blood loss worldwide. It will help in achieving the millennium development goal of reducing maternal mortality in India and globally as well. **This invention by him is NOVEL and serves a NOBEL CAUSE.**

The exceptional talent of Archit deserves due recognition. The prestigious **BAL SHAKTI PURSKAR** is the right platform. I **“highly recommend” him for the same award.**

I will be happy to share additional information about Archit and his invention.



Yours sincerely

Dr. Shyamkumar S. Sirsam

Medical Superintendent

Professor & Head of Department.

Department of Obstetrics and Gynecology

Government Medical College & Hospital

Akola.(M.S.)

Prof. & Head

Dept. of Oby. & Gyn.,

Govt. Medical College, Akola



POST PARTUM CUP

Dr. Sharad Gokhale

Professor and Dean of Infrastructure, Planning and Management
Department of Civil Engineering
Indian Institute of Technology
Guwahati – 781 039, Assam

Date: 21.07.2020

Letter of recommendation for Archit Rahul Patil

It gives me immense pleasure to recommend Master Archit Rahul Patil for the prestigious award Pradhanmantri Bal Shakti Puskar in innovative category for being so sentient about his surroundings, creative and academically excellent. His achievements in various national and international Olympiad examinations are noteworthy. His demonstrations of academic accomplishments and incredible innovativeness at such a young age are rare. He has a long list of awards and achievements to his credit, including DST INSPIRE award and Gold medal at Dr. Homibhabha Bal Vaidnyanik Science Competition.

His recent innovation of PPH cup to prevent maternal deaths is a fantastic gift to society. I therefore strongly recommend Archit for the Bal Shakti Puskar.

Prof. Sharad Gokhale

Civil Engineering Department

IIT Guwahati

Dr. Sharad Gokhale
Professor

Department of Civil Engineering
Indian Institute of Technology Guwahati
Guwahati-781039, Assam, India



POST PARTUM CUP

	
Government of Maharashtra महाराष्ट्र शासन	
Department of Obstetrics & Gynecology स्त्रीरोग व प्रसूतीशास्त्र विभाग	
Government Medical College, Aurangabad शासकीय वैद्यकीय महाविद्यालय, औरंगाबाद	
☎: (0240) 2402412-18, 22-27 (Extn. 340)	
No./OBGY/	/201
Date :	/ /201

24.7.2020

It gives me immense pleasure to issue this letter of recommendation for Master Archit Rahul Patil for the prestigious award –Pradhanmatri Bal Purskar.

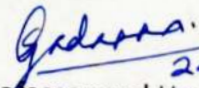
As a senior clinician and academician, my main domain of work is, advanced surgical treatment for PPH (Post partum Haemorrhage) which is not taught routinely in medical colleges. I thrive to propagate knowledge to my students and all obstetricians.

But to my pleasant surprise I have not come across a device like Archit PPH Cup. It gives the exact and real-time measurement of post delivery blood loss. It helps for early recognition and prompt treatment. It will help in reducing the maternal deaths due to delay in treatment.

This young genius boy has invented such a magic tool that is proving boon for obstetricians. He has been selected for- YOUNGEST INNOVATOR MS SIGNATURE AWARD and DST INSPIRE AWARD. He also showed academic excellence at national and international level. Recognising the talent of budding scientist will inspire the young generation to contribute to India's glory.

I strongly recommend Master Archit Rahul Patil for Bal Shakti Award for his innovation of PPH Cup.

Dr. Shrinivas N Gadappa,


24.7.2020

Professor and Head,
Department of Gynecology and Obstetrics,
Government Medical College,
Aurangabad.

प्राध्यापक व विभाग प्रमुख
स्त्रीरोग व प्रसूतीशास्त्र विभाग,
शासकीय वैद्यकीय महाविद्यालय,
औरंगाबाद.



POST PARTUM CUP



दिल्ली प्रौद्योगिकी विश्वविद्यालय
DELHI TECHNOLOGICAL UNIVERSITY
(Formerly Delhi College of Engineering)



No. DTU/T&P/2020-21/127

Dated, 27/07/2020

LETTER OF RECOMMENDATION

It gives me immense pleasure to endorse the application of "Archit Rahul Patil" who has, in my opinion, achieved great heights in the field of academia. He has shown academic aptitude and ambition beyond and above his peers. The best students are ones that genuinely love to learn. A student that pushes himself every day to learn more and be better. Archit is that kind of student. He not only performs well in academics, but also *insert co-curricular activities* Knowing more about the world, its problems and thinking about their solutions has been his passion. Following this, he has made a great invention of medical device: "post partum cup-The PPH Cup". Where the world is trying its different ways to ensure the exact estimation of blood loss, his PPH cup is such an innovation which will revolutionize the PPH management and save number of women from mortality and morbidity. He has provided the solution the problem, which even great clinicians and researchers are struggling with.

He is going to be felicitated for his incredible contribution to the society as a "YOUNGEST INNOVATOR" MS SIGNATURE AWARD an elite and prestigious award.

In the past Archit has been awarded Gold Medal for Dr. Homibhabha Bal Vaidnyanik competition for his work on sanitary waste and menstrual hygiene options.

He has shown exceptional achievements and ranking in various national and international level competitive exams conducted by SOF and Silverzone. Not only these but also single digit national ranks in exams conducted by Association of mathematics Teachers of India and Thailand Mathematics society. He has received a whopping first rank at national level in Ignited mind Lab. He is preparing hard to represent India at international junior science Olympiad- IJSO

He is a recipient of Inspire awards-Manak conducted by National Innovation Foundation NIF, Government of India.

With that being known, I recommend that a prodigy like Archit should be honoured Pradhan Mantri Bal Shakti Puraskar, for his life saving innovation and so unique record of achievements and accolades. It will encourage him to bring out the best for the betterment of the world.


(Dr. Rajesh Rohilla)
Professor
ECE Department
Delhi Technological University
Delhi -110042

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